

2023



Warner Bros. Discovery Presents the Olympic Games to Millions of Viewers Supported by Innovations Built Using AWS

Using AWS services and support, Warner Bros. Discovery broadcast the Olympic Games on its discovery+ streaming service in Europe, providing innovations for a better viewer experience.

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175 million

viewers in 50 territories across its channels and platforms



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Over 10,000

hours of audio in 19 different languages and hundreds of live events monitored using an language-checking solution



2

Increased

processing power and achieved cost savings using AWS Graviton processors

Reduced

onsite footprint and personnel

Overview

For the Olympic Games Tokyo 2020, Discovery (now [Warner Bros. Discovery](#)) required a scalable solution to broadcast the event to 50 territories across Europe in 19 languages on its broadcast channels and streaming services, including discovery+. Broadcasting and streaming the Olympic Games requires a massive effort to support delivery to viewers in multiple countries. For Tokyo 2020, 175 million viewers watched on Warner Bros. Discovery's channels and platforms, streaming 1.3 billion minutes of content, which was 21 times the amount of the previous Olympic Games. For the Olympic Winter Games Beijing 2022 that followed, the company attracted 156 million viewers and its streaming content saw 1 billion minutes consumed.

To manage an event of this scale, Warner Bros. Discovery and Amazon Web Services (AWS), worked together to develop innovations like an industry-first solution for automatically checking commentary languages. Using AWS services and [AWS Enterprise Support](#), which provides concierge-like service to achieve outcomes and find success in the cloud, the company scaled to support simultaneous video feeds and improved the viewer and commentator experience.



Opportunity | Managing Remote Broadcasting Support

Warner Bros. Discovery is a leading global entertainment and media company that operates scripted, factual and lifestyle content across film, television, and streaming. Its real-life entertainment discovery+ streaming service offers digital subscribers access to shows and exclusives, including its role as Streaming Home of the Olympics with coverage of every moment of the Games starting from Olympic Games PyeongChang 2018 and working with AWS from Tokyo 2020 into the summer of 2021.

Warner Bros. Discovery used a variety of AWS services for discovery+ and in other business areas prior to using [AWS Infrastructure Event Management](#) (AWS IEM), which offers support for planning and running business-critical events, to plan for the Olympic Games Tokyo 2020 and into the summer of 2021. Using AWS IEM, the company worked with experts from AWS before the event to strategize and during the event to address issues. Warner Bros. Discovery had support for [AWS Media Services](#) during the entire Games using [AWS Elemental Media Event Management](#) (AWS Elemental MEM), a support program designed to improve the operational reliability of business-critical video systems. Tokyo 2020 and Beijing 2022 were a global effort with support teams and expertise sourced across continents in Asia, the United States, and Europe 24 hours a day. “During these major events, we had important AWS IEM and AWS Elemental MEM teams supporting us and we could count on this support,” says Ian Cockett, Group Vice President, Engineering Technology at Warner Bros. Discovery. “When we needed it, all the AWS experts around the world were on hand to help overcome any difficulties we had.”

Because of understandable on-the-ground and travel restrictions related to the COVID-19 pandemic, Warner Bros. Discovery delivered the Games with a greatly reduced technical and personnel footprint for Tokyo 2020 compared to its previous Games experience. Given the on-going situation, the company further reduced its onsite presence for Beijing 2022, delivering the same high quality coverage and programming using support that included AWS IEM and AWS Elemental MEM and AWS services like [AWS Elemental MediaLive](#), a broadcast-grade live video processing service to encode live video for broadcast and streaming to any device. “The company’s remote production journey was well underway prior to this becoming more essential during the pandemic. As we look to the future, it is a huge benefit to know we have the ability to send only the right and required people to major events, such as the Olympics, because our in-house technology and expertise complimented by AWS services in the cloud has evolved so much. This has the clear staffing, business, and sustainability benefits both we and our partners are seeking” says Ian Cockett.

Solution | Providing Scalability and Flexibility

Scalability was essential for the Olympic Games coverage because Warner Bros. Discovery needed a solution that could stream dozens of simultaneous video feeds. To manage this scale, the company used [Amazon Elastic Compute Cloud](#) (Amazon EC2), which offers secure, resizable compute capacity for virtually any workload. Additionally, because live sports can have weather delays or run long, Warner Bros. Discovery needed the flexibility to scale based on immediate needs rather than a schedule. The company achieved operational cost savings and increased processing power by using [AWS Graviton processors](#), which are designed to deliver the best price performance for cloud workloads running in Amazon EC2.

Warner Bros. Discovery also worked alongside [AWS Professional Services](#), which provides support for achieving desired business outcomes using AWS services, to design and implement an innovative Digital Audio Commentary Checker that checks that the right language is broadcast to the right country using a serverless microservices architecture to scale the solution without manual intervention. “As a multi-market broadcaster across Europe, managing such a diversity and sizable number of languages is one of the most challenging projects we undertake,” says Ian Cockett. “Getting the right commentary on the right sport at the right time is clearly vital and complex, especially during the pandemic when many commentators were required to conduct their roles remotely”. The industry-first solution addresses a significant challenge for broadcasting live sporting events by using automatic speech recognition machine learning and [Amazon Transcribe](#), which automatically converts speech to text, to identify live commentary languages and automatically notify operational teams of audio issues in near real time. On average, Warner Bros. Discovery broadcasts 61 concurrent live events in up to 19 different languages, making manual checks time consuming and difficult. The automated solution saves time by complimenting manual checks and flagging potential issues much quicker. After introducing the solution for Tokyo 2020, the company has used it to monitor all major live events such as tennis Grand Slam tournaments and cycling’s Grand Tours.

For commentators, Warner Bros. Discovery created a system with background data about the athletes using [Amazon WorkSpaces](#), a fully managed desktop virtualization service that provides secure, reliable, and scalable access to persistent desktops from any location. Commentators can access the system from all over the world, which was critical for the additional remote support needed to meet remote requirements during the pandemic.

Outcome | Innovating for Future Olympics

Warner Bros. Discovery is well on the road to Olympic Games Paris 2024 where it has plans to continue to enhance the viewing experience and innovate around how audiences engage with the Games. As the company is doing around a number of other major events, it plans to build on the experience of employing AWS IEM and AWS Elemental MEM, as well as continue to collaborate with AWS to support this development and innovation. “The ‘AWS’ team from AWS IEM and AWS Elemental MEM slot in seamlessly within our team, and more than meet our requirements from a technical team or cloud provider,” says Ian Cockett. “You know support is available when you need it and the AWS team also identifies possible issues beforehand to get them addressed.”

About Warner Bros. Discovery

Warner Bros. Discovery is a leading global media and entertainment company that creates and distributes the world’s most differentiated and complete portfolio of content and brands across television, film and streaming. Available in more than 220 countries and territories and 50 languages, Warner Bros. Discovery entertains audiences worldwide through its iconic brands and products. Its discovery+ streami



digital subscribers access to real-life entertainment programming and sports, including as the streaming home of the Olympic Games in Europe.

AWS Services Used

AWS Elemental MediaLive

AWS Elemental MediaLive is a broadcast-grade live video processing service that creates high-quality streams for delivery to broadcast TVs and internet-connected devices.

[Learn more »](#)

AWS Graviton processors

AWS Graviton is a family of processors designed to deliver the best price performance for your cloud workloads running in Amazon Elastic Compute Cloud (Amazon EC2).

[Learn more »](#)

Amazon Transcribe

Automatically convert speech to text.

[Learn more »](#)

Amazon WorkSpaces

Fully managed, secure, reliable virtual desktops for every workload.

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Ringier Axel Springer Polska (RASP) drastically reduced its carbon emissions after migrating to AWS. RASP, a Polish digital media company, committed to reducing carbon emissions by 90 percent. As part of a strategy geared toward compute efficiency and price performance, it migrated to Amazon EC2 Graviton processors, which cut its carbon emissions for IT systems to near



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